

A Scalable Matheuristic for Routing Capacity-Constrained Groundfish Surveys

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Abstract. Bottom trawl groundfish surveys are planned around a fixed set of sampling stations, but the order of visits and the timing of port returns strongly affect total travel distance and operational feasibility. Although mixed-integer programming (MIP) models can encode capacity and port-call constraints directly, solving the full model to optimality is often impractical at realistic scales. We propose a matheuristic that combines fast tour construction with repeated calls to a time-limited capacity MIP subproblem. Starting from an initial segmented tour, the algorithm iteratively examines boundaries between adjacent segments separated by a port visit and solves a restricted two-segment capacity MIP on the union of stations in the two segments, with fixed endpoints and a single intermediate port call. The MIP reallocates stations between the two segments subject to capacity, and the best incumbent found within a time limit is accepted if it improves total distance. We evaluate the approach under four initialization strategies and a range of MIP time limits. Across scenarios, the method consistently improves baseline plans and provides a fast, reproducible tool for strategic planning and scenario comparison. These results suggest that restricted-MIP local improvement is an effective mechanism for producing high-quality survey routes under realistic operational constraints.

Keywords: Groundfish Survey Problem · Multi-trip VRP · Capacitated Routing with Replenishment · Segment-Based Decomposition · Matheuristics

1 Introduction

Bottom trawl groundfish surveys require visiting a set of sampling stations while respecting operational constraints such as limited onboard capacity, time, and cost. Even when station locations are fixed, the order of visits and the timing of port returns directly influence survey feasibility and efficiency. A method that can generate good survey routes quickly is therefore valuable for both survey planning and survey design.

From an optimization perspective, these decisions couple routing and capacity in a way that makes exact mixed-integer programming (MIP) approaches computationally challenging at realistic scales. Matheuristics address this difficulty by combining mathematical programming with heuristic search [1], often using MIP subproblems to improve a larger solution [2].

This idea appears in several forms in the literature. Fan et al. use a MIP model as a local search operator, repeatedly re-optimizing a selected subset of decisions [5]. Dell’Amico et al. follow a similar strategy, solving MIP subproblems within a heuristic loop and, when needed, imposing a time limit and accepting the best incumbent returned by the solver [4]. This use of time-limited MIP calls—leveraging the best solution found rather than requiring full optimality—is a key ingredient shared with our approach.

Building on these ideas, we introduce a scalable matheuristic for routing capacity-constrained groundfish surveys. Starting from a feasible segmented route, the method applies short, targeted MIP refinements to adjust segment boundaries under vessel capacity. Improvements are accepted whenever found, and the process repeats until no further gains arise. The approach provides a fast and practical baseline planning tool for survey design and scenario evaluation.

Section 2 describes the survey-routing problem, and section 3 presents the capacity-aware MIP model. Because solving this full end-to-end MIP is computationally prohibitive at operational scale, section 4 introduces the scalable matheuristic that effectively addresses the problem. Finally, sections 5 and 6 report the experimental setup and results, followed by discussion and conclusions.

2 Groundfish survey problem

Groundfish surveys provide critical information on the abundance, distribution, and biological characteristics of fish stocks and serve as the empirical foundation for sustainable harvest management. In Iceland, these surveys have followed a fixed-station sampling design since 1985, with the same set of trawl stations revisited annually in the spring to maintain a consistent long-term time series [7]. This design aligns with established statistical principles for monitoring population trends [10], while also reflecting the operational continuity of the Icelandic survey program.

For a survey with a given set of stations, the survey leader must determine an efficient routing plan for one or more vessels tasked with visiting and towing at each station. This routing problem is shaped by regulatory and operational constraints. During the survey, some of the catch is retained in order to be sold upon landing. As a result, vessels must periodically return to designated ports to unload because each vessel’s catch capacity is limited. In practice, unloading occurs at regular intervals (typically within five days of operations), embedding mandatory port calls into the routing plan.

The amount of fish caught at each station is uncertain and can vary substantially from year to year. Therefore, routing plans must be robust to variability in catch volumes, ensuring that capacity constraints are not violated while still

maintaining survey coverage. When multiple vessels operate simultaneously, the stations must be partitioned between them in a way that respects their capacities and schedules.

Each trawl station is defined by two tow endpoints, and feasible tow directions may be limited by local bathymetry, prevailing weather, or vessel-specific operational constraints. Consequently, travel between stations is directional: distances from station A to station B may differ from those from B to A. Naïve point-to-point paths can also cross land; to prevent this, we compute waypoint-aware distances that reroute any invalid land-crossing arcs through intermediate waypoints. These corrected, direction-aware distances serve as inputs to both the MIP model and the matheuristic developed for large-scale routing.

Prior research has examined this routing task through both early genetic-algorithm formulations of the Groundfish Survey Problem [9] and more recent interactive evolutionary reoptimization approaches for dynamic survey planning [13].

In general, the problem involves the following. (i) a large number of geographically distributed stations, (ii) direction-dependent travel and towing constraints and (iii) regulatory requirements that impose periodic port returns due to mandatory retention and finite vessel capacity.

3 Capacity-Aware MIP Model

The survey-routing problem shares features with multi-trip vehicle routing problems (VRP), since capacity limitations force repeated returns to port. At the same time, it differs from classical VRP formulations because all stations must be visited exactly once and the solution forms a single directed tour with capacity-induced segment breaks. In effect, the problem is a hybrid between a multi-trip VRP and a directed traveling salesman problem (TSP) with mandatory unloads.

Node representation. We represent the survey as a directed tour over a graph whose nodes include vessel, port and trawl station start- and endpoint. Locations are indexed by $i \in \{0, \dots, n'\}$, where n' denotes the total number of paired endpoint nodes for the vessel, stations, and ports. Location $i = 0$ is the vessel, and each trawl station $i \in S = \{1, \dots, n\}$ is modeled by a pair of endpoints $(\text{lat}_{2i}, \text{lon}_{2i})$ and $(\text{lat}_{2i+1}, \text{lon}_{2i+1})$ so that the model can explicitly choose the tow direction. Here, $d_{i,j}$ denotes the waypoint-aware travel distance between nodes i and j . The vessel is represented by nodes 0 and 1 with $d_{0,1} = d_{1,0} = 0$, which fixes the tour to start at node 0 and end at node 1 with zero internal cost. Ports form the set P , indexed as $k \in \{n+1, \dots, n'\}$, and each port is also represented by two nodes $(2k, 2k+1)$ with $d_{2k,2k+1} = 0$ for consistency. Let $x_{i,j} \in \{0, 1\}$ indicate whether the tour uses arc (i, j) . For a given vessel we have a predetermined capacity C and the start/end node pair.

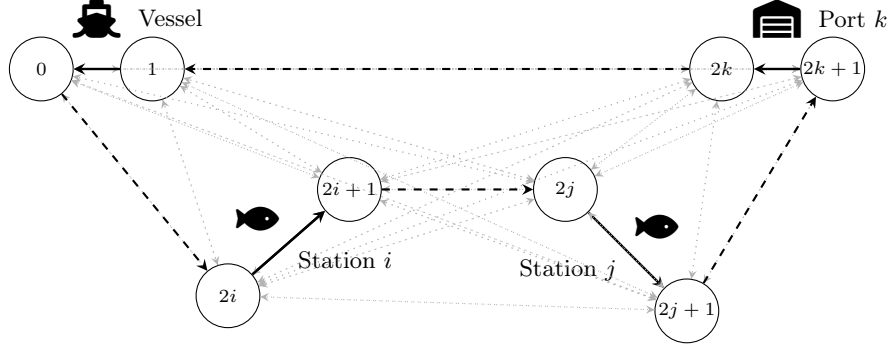


Fig. 1: Indexed node notation used in the MIP model. The underlying structure is a fully bidirectional graph: all node pairs are connected by dashed gray edges to indicate every feasible arc (a, b) in the waypoint-aware distance matrix. Solid black arrows represent mandatory internal arcs that enforce movement from each start node to its corresponding paired end node (e.g., $2i \rightarrow 2i+1$), capturing the directed towing or port-reset structure. Dashed black arrows show an example tour demonstrating how the vessel may travel from the boat start, through a sequence of stations, then to a port, and finally back to the boat end node.

Degree constraints. To ensure the solution forms a single tour that visits every node exactly once, each node has one incoming and one outgoing arc:

$$\begin{aligned} \sum_{i \in \{0, \dots, 2n'+1\}} x_{i,j} &= 1 & \forall j \in \{0, \dots, 2n'+1\} & \quad (\text{in-degree}) \\ \sum_{i \in \{0, \dots, 2n'+1\}} x_{j,i} &= 1 & \forall j \in \{0, \dots, 2n'+1\} & \quad (\text{out-degree}) \end{aligned}$$

Because ports are modeled using the same two-endpoint construction as stations, these constraints imply that each selected port is also visited exactly once.

Implications for port selection. Since any port included in the MIP must be visited exactly once, the model cannot choose ports dynamically. We therefore fix the visited-port set (P_{fix}) and use duplicated port nodes to encode repeated calls. The matheuristic described later avoids this limitation.

Direction choice. For each paired endpoint $(2i, 2i+1)$ we enforce

$$x_{2i,2i+1} + x_{2i+1,2i} = 1. \quad (\text{pair})$$

This selects a single internal towing direction for each station (and port) pair.

Objective. We minimize total travel distance over the selected arcs:

$$\min \sum_{\substack{i \in \{0, \dots, 2n'+1\} \\ j \in \{0, \dots, 2n'+1\}}} d_{i,j} x_{i,j}. \quad (\text{objective})$$

Subtour elimination. Without connectivity constraints, the tour can split into subtours. We add cuts via callbacks: for any subtour arc set $\mathcal{P}$,

$$\sum_{(i,j) \in \mathcal{P}} x_{i,j} \leq |\mathcal{P}| - 1. \quad (\text{subtour elimination})$$

This prevents disconnected cycles and yields a single connected tour.

Capacity and catch flow. To enforce capacity and represent cumulative catch carried along the tour, we use directed flow variables $w_{i,j}$ for the load on arc (i,j) , and $v_{i,j}$ for catch injected at station endpoints on the internal arc. Before presenting constraints, note that $w_{i,j}$ must respect both direction and capacity and propagates the load forward along selected arcs.

Flow conservation at stations. For each station $j \in \{1, \dots, n\}$, we enforce conservation at both endpoints:

$$\begin{aligned} \sum_{k \in \{0, \dots, 2n'+1\} \setminus \{2j+1\}} w_{k,2j} + v_{2j,2j+1} &= w_{2j,2j+1}, \\ \sum_{k \in \{0, \dots, 2n'+1\} \setminus \{2j\}} w_{k,2j+1} + v_{2j+1,2j} &= w_{2j+1,2j}. \end{aligned} \quad (\text{flow conservation})$$

The station catch injected on the internal arc satisfies

$$v_{2j,2j+1} + v_{2j+1,2j} = a_j. \quad (\text{station expected catch})$$

Flow transfer after towing. After completing the tow, the accumulated load must transfer to the next arc in the tour:

$$\begin{aligned} w_{2j,2j+1} &= \sum_{k \in \{0, \dots, 2n'+1\} \setminus \{2j,2j+1\}} w_{2j+1,k}, \\ w_{2j+1,2j} &= \sum_{k \in \{0, \dots, 2n'+1\} \setminus \{2j,2j+1\}} w_{2j,k}. \end{aligned} \quad (\text{flow transfer})$$

These equalities propagate the accumulated catch from a station to the next arc.

Port unload. At ports, the vessel unloads before leaving. For each port $k \in \{n+1, \dots, n'\}$ and each of its nodes $p \in \{2k, 2k+1\}$:

$$\sum_{j \in \{0, \dots, 2n'+1\}} w_{p,j} = 0. \quad (\text{port reset})$$

Capacity bounds. Capacity is enforced by

$$w_{i,j} \leq C x_{i,j}, \quad v_{i,j} \leq C x_{i,j}. \quad (\text{capacity})$$

Together with the port reset constraints, this forces a return to port once the accumulated catch approaches capacity.

Algorithm 1: General capacity aware MIP (**C-MIP**) procedure for computing a feasible end-to-end survey tour.

Input: *Problem instance:* stations S , fixed port set P_{fix} , boat capacity C , distances $d_{i,j}$. *Solver controls:* time limit L

Output: Best feasible tour found within L

- 1 Build the directed graph with paired nodes for the vessel ($i = 0$), stations ($i \in S$), and ports ($i \in P_{\text{fix}}$);
 - 2 Duplicate port nodes to encode the fixed maximum number of port calls;
 - 3 Create the MIP with degree, pair, flow, and capacity constraints;
 - 4 Enable subtour elimination via callback;
 - 5 Set solver time limit to L ;
 - 6 Optimize and record the best feasible incumbent;
 - 7 **return** incumbent tour and objective (if any);
-

Discussion and complexity. The full formulation presented here, which we refer to as the C-MIP (capacity-aware MIP), extends classical flow-based subtour-elimination models for the TSP [3] by incorporating directed towing, load propagation, mandatory port resets, and a vessel capacity constraint. These additions substantially increase computational difficulty, especially when port choices and the number of required port calls are not fixed in advance. The no-port variant (NP-MIP) removes all capacity constraints, reducing the problem to the classical directed TSP. Algorithm 1 summarizes how the full C-MIP is instantiated and solved.

4 Matheuristic Framework

We model a tour as an ordered list of capacity-feasible segments separated by ports. Let C denote vessel capacity. Each segment is a directed path that begins at a port (or the vessel’s start location), visits a subset of stations whose accumulated, expected catch remains below C , and ends at the next port where the vessel unloads. The *capacity-aware* matheuristic (MH) constructs such a tour by first producing an initial station ordering, then forming an initial capacity-feasible segmentation, and finally iteratively improving it by re-optimizing *adjacent* segment pairs using a C-MIP for this subproblem. This yields a scalable approach that exploits the problem’s structure while avoiding the combinatorial difficulty of solving the full C-MIP.

Algorithm overview. The full matheuristic consists of two phases: an initialization phase that produces an initial ordering and segmentation, and an iterative boundary improvement phase that repeatedly refines adjacent segment pairs. The overall procedure is given in algorithm 2. Whenever a boundary is selected during the sweep, the algorithm invokes the local two-segment optimization routine in algorithm 3. Accepted changes update the segmentation and trigger local re-optimization of the affected segments. The process terminates once a full sweep yields no improvements.

Algorithm 2: Matheuristic Boundary-Sweep Framework (MH)

Input: Stations S , ports P , capacity C , waypoint-aware distances $d_{i,j}$, start/end vessel nodes, initialization mode `InitMode`, single-segment TSP limit $L_{1\text{seg}}$, two-segment C-MIP limit $L_{2\text{seg}}$

Output: Best segmented tour found

- 1 **Phase 0: Initialization and baseline;**
- 2 Obtain S_{ord} and initial partition $\mathcal{V}$ using `InitMode`;
- 3 Evaluate each segment in $\mathcal{V}$ using the directed segment-TSP (limit $L_{1\text{seg}}$);
- 4 Mark all boundaries as *active*;
- 5 **Phase 1: Boundary-sweep improvement;**
// A sweep corresponds to a full traversal of all boundaries in circular order.
- 6 **repeat**
- 7 **foreach** boundary b between adjacent segments in tour order **do**
- 8 **if** b is not active **then**
- 9 | continue
- 10 **end**
- 11 Mark b as inactive;
- 12 Call algorithm 3 on boundary b (using limits $L_{1\text{seg}}$ and $L_{2\text{seg}}$);
- 13 **if** boundary changed **then**
- 14 | Update $\mathcal{V}$;
- 15 | Mark b and any other modified segments as *active*;
- 16 **end**
- 17 **end**
- 18 **until** no boundary changed in a full sweep;

Initial ordering. The framework requires an initial ordering of the survey stations from which a capacity-feasible tour can be constructed. This ordering may be produced by any heuristic or optimization-based procedure, and the choice can substantially influence the resulting segmentation and solution quality. A wide variety of initialization strategies are available in practice, offering different balances between computational effort and ordering quality. In our experimental study (see section 5), we examine four representative initialization variants ranging from lightweight greedy methods to a more computationally demanding NP optimum based baseline.

Initial segmentation. Let S_{ord} denote the ordered list of stations. From the total expected catch, we compute the minimum number of required segments $N_{\min} = \lceil (\sum_{j \in S} a_j) / C \rceil$. The algorithm then scans the ordering, accumulating load and inserting the nearest feasible port (with respect to $d_{i,j}$) either *before* a station if adding it would exceed capacity, or *after* a station once the accumulated load reaches C and additional stations remain. Each port insertion resets the load.

The resulting visit list $\mathcal{V} = \{S_1, \dots, S_N\}$ defines a capacity-feasible segmentation with $N \geq N_{\min}$, forming a partition of the station set. Each segment is then solved as a directed segment-TSP (time limit $L_{1\text{seg}}$), producing a baseline segmented tour. All segment boundaries are marked as eligible for improvement.

Algorithm 3: Optimize Boundary Capacity (C-MIP subroutine in MH)

Input: Boundary b between segments S_L and S_R , boundary port p , capacity C , single-segment TSP limit $L_{1\text{seg}}$, two-segment C-MIP limit $L_{2\text{seg}}$

Output: Updated segments S'_L, S'_R and, if fallback is used, the updated donor segment S'_D ; otherwise no change

- 1 Set $s_D = \emptyset$ (donor station);
- 2 **Boundary setup:** Set $S_{2\text{seg}} = S_L \cup S_R \cup \{s_D\}$, where s_D (when used) is drawn from some *other* neighboring segment S_D and set $S'_D = S_D \setminus \{s_D\}$;
- 3 Fix start and end nodes around $S_{2\text{seg}}$ and enforce exactly one visit to p ;
- 4 Solve the two-segment C-MIP (algorithm 1) on $S_{2\text{seg}}$ with time limit $L_{2\text{seg}}$;
- 5 Extract tentative segments S'_L and S'_R from the resulting C-MIP tour;
- 6 Re-optimize each tentative segment using a directed segment-TSP with $L_{1\text{seg}}$;
- 7 **if** *either segment-TSP fails* **then**
- 8 | **return** no change;
- 9 **end**
- 10 **if** $d(S'_L) + d(S'_R) < d(S_L) + d(S_R)$ **then** two-segment improvement only
- 11 | **return** updated segments S'_L, S'_R
- 12 **else if** $d(S'_L) + d(S'_R) + d(S'_D) < d(S_L) + d(S_R) + d(S_D)$ **then** fallback used
- 13 | **return** updated segments S'_L, S'_R, S'_D
- 14 **else if** $s_D = \emptyset$ **then** *fallback strategy*
- 15 | Select the closest station s_D from any donor segment $S_D \in \mathcal{V} \setminus \{S_L, S_R\}$ such that inserting s_D into the nearer of $\{S_L, S_R\}$ remains capacity-feasible;
- 16 | Go back to **boundary setup**;
- 17 **end**
- 18 **return** no change;

Two-segment subproblem. When the sweep selects a boundary b between adjacent segments S_L and S_R with separating port p , we solve a local C-MIP on $S_L \cup S_R$ with the start node fixed (previous port or vessel start), the end node fixed (next port or vessel end), and a single port visit forced at p . The C-MIP attempts to reassign stations between the two segments subject to capacity. A time limit $L_{2\text{seg}}$ is imposed and the best incumbent is used; if either resulting segment-TSP fails, the change is rejected. If this local C-MIP produces no accepted change, a fallback strategy is attempted that selects the station outside $\{S_L, S_R\}$ with minimum distance to either segment, inserts it into the nearer side (if capacity allows), and re-optimizes the left, right, and donor segments. The move is accepted only if the summed distance of these three segments strictly decreases. The detailed C-MIP procedure is given in algorithm 3.

Sweep and termination. The algorithm sweeps boundaries in circular tour order. A boundary is defined between adjacent segments S_k and $S_{k+1 \bmod N}$, and whenever such a boundary is modified, its two neighboring boundaries (at $k-1 \bmod N$ and $k+1 \bmod N$) are marked *active* for re-evaluation. A *sweep* consists of visiting all boundaries once in this circular order. The algorithm terminates when a full sweep yields no accepted changes.

5 Experimental Setup

Data and instance summary. The data come from the 2023 spring groundfish survey conducted by the Icelandic Marine and Freshwater Research Institute [8]. The instance contains $n = 580$ trawling stations and $|P| = 23$ ports, with 7 repeated port visits. The full dataset also includes 27 waypoint nodes and 4 survey vessels. For simplicity, all experiments consider a single vessel, Árni Friðriksson, based in Hafnarfjörður (fixing $(\text{lat}_0, \text{lon}_0)$ and $(\text{lat}_1, \text{lon}_1)$) with capacity $C = 45$ tonnes. In total, the model uses 611 paired endpoint nodes (stations, ports, and vessel start/end). Station catch values a_j are expected catches derived from historical time series. The total catch is 528 tonnes, implying a minimum of $N_{\min} = 12$ capacity-feasible segments. Distances $d_{i,j}$ are waypoint-aware shortest-path distances that reroute land-crossing arcs via waypoint nodes.

Port configuration. The C-MIP uses a fixed port list P_{fix} , which includes duplicated port entries to represent repeated visits and is set here to match the port structure of the 2023 survey, and requires $N = 13$ segments. In contrast, the matheuristic selects ports dynamically from the unique candidate set P .

Experimental scenarios. We evaluate three model classes. The NP-MIP is a no-port directed TSP over all stations and provides both an optimal port-free routing and a theoretical lower bound on achievable distance. The C-MIP is the full capacitated MIP using the fixed port set P_{fix} (with duplicated nodes to encode repeated visits). Finally, the MH variants implement the proposed matheuristic, which applies the C-MIP only to small two-segment subproblems and differs solely in the initialization strategy used to construct the initial station ordering. All experimental configurations are summarized in table 1.

Initialization strategies. The MH experiments consider four alternative constructions of the initial station ordering. The NN (nearest-neighbor) strategy repeatedly selects the closest feasible unvisited station, inserting ports when capacity is exceeded; it is extremely fast but can produce noisy segmentations. The CI (cheapest insertion) strategy grows a tour by inserting at each step the station that causes the smallest marginal increase in the length of tour that results in smoother orderings while remaining lightweight [12]. The GE (greedy edge) strategy follows a Kruskal-like process, adding the shortest admissible edges while avoiding premature subtours before inserting ports [11]; it is often competitive but sometimes less stable than CI. Finally, the OPT initialization derives an optimal no-port directed TSP (via NP-MIP) and then inserts ports to satisfy capacity, giving high-quality orderings at higher computational cost.

MIP and solver parameters. All models are solved with Gurobi 13 [6]. Even with a long time limit ($L_{\text{cap}} = 12$ hours), the C-MIP on the full dataset gives no feasible solution. In contrast, the baseline NP-MIP solves in just over an hour without imposing a limit on L_{NP} . We likewise leave the single-segment directed TSP subproblems uncapped ($L_{1\text{seg}}$); these typically solve to optimality

within seconds (only occasionally requiring up to a few minutes) and therefore do not constitute a computational bottleneck. Instead, the two-segment C-MIP subproblem is the primary bottleneck, so we evaluate a range of limits $L_{2\text{seg}} \in \{1, 2, \dots, 8\}$ minutes to assess effects on solution quality and robustness.

Matheuristic parameters. The boundary-sweep procedure is capped at 100 full sweeps, which is ample for all runs. Boundary updates consider only the immediate neighbors (i.e., radius 1 in each direction), although larger neighborhoods would also be possible. After each accepted change, the two affected segments are re-solved via the directed segment-TSP. All time limits for segment subproblems follow the settings given above.

6 Results

Baseline initialization quality. Before evaluating MH performance, table 1 reports the baseline distances produced by each initialization strategy. The four strategies yield markedly different starting qualities: NN produces the weakest port-feasible route (6911.67 nm), CI and GE give considerably stronger orderings, and the OPT initialization—obtained by solving the no-port directed TSP—achieves the strongest baseline (4762.72 nm) at the expense of a 59-minute preprocessing step. All strategies produce $N = 13$ capacity-feasible segments. These differences frame the improvement potential under MH refinement.

MH performance under two-segment time limits. Table 2 reports the sensitivity of the MH runs to the per-call two-segment C-MIP time limit $L_{2\text{seg}}$ for all four initialization strategies (NN, CI, GE, OPT). For each strategy, we report the number of boundary-sweep passes, the final total distance (nm), the mean/median/maximum number of stations that moved *within their current segments* per sweep, the number of accepted vs. total two-segment solves, the resulting C-MIP optimality-gap statistics (min/avg/max/std, %), and the total wall-clock runtime (minutes).

Table 1: Variant notation, solver limits, and its baseline initialization distances (nm). The first distance column gives the underlying no-port initialization, and the second shows the corresponding capacity-feasible initialization with ports. Some variants do not entail either a no-port or a with-port solution.

Notation	Variant	Solver limits	No port	With port
NP-MIP	No-Port directed TSP	$L_{\text{NP}} = 2$ hrs	3562.57	N/A
C-MIP	Capacity-aware MIP	$L_{\text{cap}} = 12$ hrs	N/A	(timeout)
MH-NN	MH with nearest-neighbor	$L_{2\text{seg}} \in 1\text{-}8$ min	N/A	6911.67
MH-CI	MH with cheapest-insertion	$L_{2\text{seg}} \in 1\text{-}8$ min	4377.20	5354.93
MH-GE	MH with greedy-edge	$L_{2\text{seg}} \in 1\text{-}8$ min	5027.03	5104.34
MH-OPT	MH with NP-based	$L_{2\text{seg}} \in 1\text{-}8$ min (NP-MIP)	4762.17	

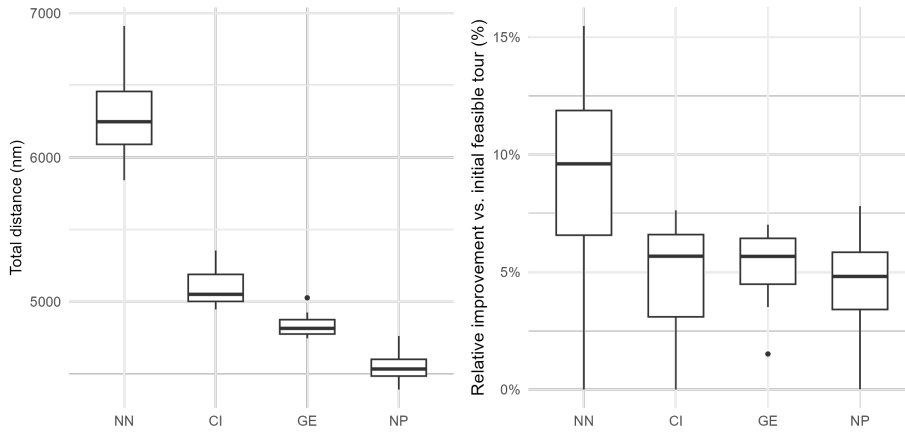


Fig. 2: Performance of the matheuristic under four initialization strategies. Left: final route distances. Right: relative improvements over its initial route.

Despite the use of time-limited C-MIP solves—where $L_{2\text{seg}}$ dominates computational cost—repeated sweeps with segment re-optimization consistently improve the global tour beyond the initial segmentation.

Two consistent patterns emerge across all strategies. First, moderate limits (approximately 2–3 min) tend to yield the best final distances for a given initialization, whereas substantially larger limits (4–8 min) increase runtime without producing systematic gains. Second, both the fraction of accepted solves and the typical intra-segment movement per sweep remain broadly stable across limits, indicating that many short local solves are as effective and more time-efficient than fewer long solves.

Patterns of improvement. The distribution of final distances and relative improvements is visualized in fig. 2. NN yields both the longest initial and the longest final routes, yet it achieves the largest relative gain under MH refinement. In contrast, GE and OPT begin from much stronger orderings and therefore show smaller—but far more stable—relative improvements, ultimately producing the best final routes. These patterns confirm that the MH approach consistently improves every baseline, while higher-quality initializations retain an advantage in the final objective.

Representative improvements. Figure 3 compares the initial capacity-feasible route derived from the NP-MIP ordering with the final refined route produced by the matheuristic (i.e. MH-OPT). These examples show how localized segment-boundary changes propagate into meaningful whole-route improvements. In line with broader trends, improvements often arise near large catch jumps, where capacity constraints force port returns and segment boundaries become influential.

Table 2: MH performance by initialization strategy under varying two-segment C-MIP time limits.

(a) MH-NN: Matheuristic using nearest neighbor initialization

$L_{2\text{seg}}$ (min)	num. sweeps	final distance	stations moved mean/med/max	accepted/ solves min	MIP gap (%) mean/max/std	run time (min)
1	5	6561.790	5.4/4/11	9/39	0/15/46/10	118
2	6	6090.243	5.3/4/11	14/47	0/18/64/13	245
3	5	6051.153	5.6/6/12	11/39	0/11/24/6	285
4	4	6688.998	4.8/3.5/12	6/30	0/15/84/16	302
5	9	6012.539	4.0/3/12	14/54	0/12/46/9	699
6	6	6252.239	6.0/6.5/11	19/50	0/9/40/10	781
7	6	6163.591	4.8/4.5/11	14/43	0/10/30/8	828
8	5	6218.146	6.4/6/11	13/48	0/10/34/8	1063

(b) MH-CI: Matheuristic using cheapest insertion initialization

$L_{2\text{seg}}$ (min)	num. sweeps	final distance	stations moved mean/med/max	accepted/ solves min	MIP gap (%) mean/max/std	run time (min)
1	2	5000.347	5.5/5.5/11	4/24	0/14/99/20	66
2	2	5108.418	4.5/4.5/9	3/22	0/6/18/6	86
3	3	5062.339	4.0/2/10	4/24	0/7/18/6	155
4	5	5012.521	3.8/3/10	7/33	0/6/17/5	267
5	3	5067.459	5.3/6/10	7/28	0/5/14/5	247
6	5	4886.220	3.4/2/9	8/32	0/3/14/4	365
7	3	5108.968	3.7/3/8	4/24	0/5/13/4	367
8	4	5025.292	3.5/3/8	7/27	0/5/13/4	485

(c) MH-GE: Matheuristic using greedy edge initialization

$L_{2\text{seg}}$ (min)	num. sweeps	final distance	stations moved mean/med/max	accepted/ solves min	MIP gap (%) mean/max/std	run time (min)
1	6	4799.879	3.5/2.5/9	12/39	0/12/37/9	109
2	5	4751.567	4.6/5/8	12/40	0/11/52/10	229
3	6	4771.586	4.7/3.5/10	16/44	0/8/25/7	313
4	3	4850.068	4/4/8	7/27	0/8/29/7	267
5	5	4773.525	3.8/3/9	11/37	0/7/36/7	455
6	6	4853.620	3.8/3.5/9	16/39	0/6/26/6	549
7	3	4813.738	5.7/6/8	8/31	0/5/26/5	504
8	6	4746.185	3.5/2/9	11/39	0/6/20/5	825

(d) MH-OPT: Matheuristic using an optimal NP-MIP initialization

$L_{2\text{seg}}$ (min)	num. sweeps	final distance	stations moved mean/med/max	accepted/ solves min	MIP gap (%) mean/max/std	run time (min)
1	3	4533.433	3.7/4/7	7/23	0/14/99/21	89
2	3	4592.349	3.0/2/7	5/21	0/11/26/7	117
3	6	4484.448	4.7/5.5/8	10/48	0/9/34/7	388
4	2	4600.363	2.5/2.5/5	3/16	0/7/17/6	171
5	4	4437.978	4.5/5/8	8/31	0/8/62/12	312
6	5	4500.298	3.8/3/8	8/33	0/7/19/7	396
7	2	4604.483	2.5/2.5/5	3/16	0/6/14/5	285
8	10	4390.774	3.5/3/7	10/64	0/8/30/6	1120

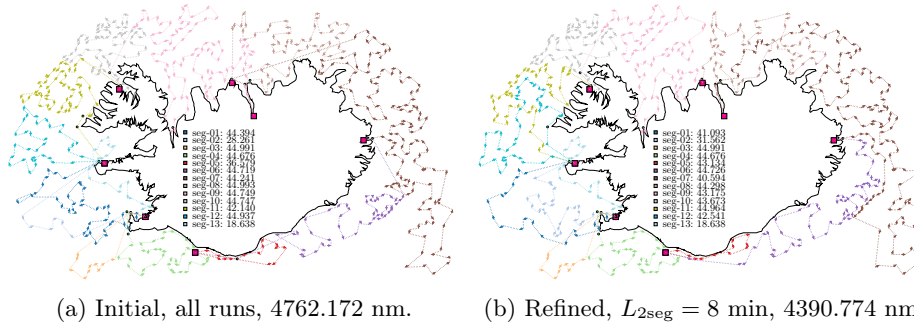


Fig. 3: Representative example showing the initial capacity-feasible route (a) and the refined route produced by the matheuristic (b), using the OPT initialization.

Best observed performance. Across all runs, the best final plan improves the baseline distance from 4762.172 nm to 4390.774 nm, a reduction of approximately 270 nm (5.8%). At realistic steaming speeds (8-12 kn), this corresponds to roughly 23-34 hours saved—about a full day at sea.

Time-limit sensitivity. The two-segment C-MIP subproblem is the primary computational bottleneck. It typically identifies a feasible incumbent within 1-8 minutes, but increasing the per-call time limit beyond 3 minutes yields no systematic improvement in the final objective (see table 2), while significantly increasing runtime. Thus, short, repeated local MIP solves are more effective—and more computationally efficient—than long single solves.

7 Discussion

The MH acts as a strong local improvement operator: it consistently upgrades diverse initializations, yet benefits most when given a well-structured starting route. This explains why strong initializations retain an absolute advantage even as weaker ones enjoy larger relative gains—MH is not designed to redesign the global station order, but to make locally optimal boundary adjustments.

Computationally, the two-segment C-MIP delivers high-quality incumbents quickly, enabling effective progress under short per-call limits, while longer limits mainly inflate runtime. This favors many short local solves over a few long ones in operational settings.

Methodologically, the segmentation membership is only weakly adjustable (via limited boundary augmentation), so the search can converge to high-quality local optima under the current neighborhood. Broadening controlled ways to transfer stations across boundaries is the most direct path to further gains.

Operationally, the approach produces fast, reproducible plans suitable for annual survey planning and scenario analysis, improving current deterministic baselines without requiring intractable end-to-end optimization.

8 Future Work

The current MH optimizes repeated two-segment neighborhoods, which limits global exploration and precludes guarantees of global optimality. Future extensions should broaden the neighborhood while controlling runtime—e.g., occasional three-segment optimizations, controlled station transfers across multiple boundaries, and periodic “shake” moves that resegment a larger region before resuming two-segment improvement.

A natural next step is a multi-vessel extension in which several boats cooperatively service the station set. In that setting, distance alone is insufficient: incorporating station service times shifts the emphasis toward balancing completion times (e.g., minimizing makespan) while respecting vessel-specific capacities and time-at-sea limits. A promising approach is a two-level matheuristic that (i) constructs a global backbone ordering and (ii) partitions stations among vessels with local boundary refinements; this readily accommodates heterogeneous fleets with different speeds, capacities, and operating limits.

Uncertainty is another practical priority. Sailing times, service times, and realized catches vary widely year to year, whereas the current MH assumes deterministic expected catches. A straightforward extension is to assess feasibility and performance via empirical sampling from historical distributions of catches and service times, favoring solutions that remain feasible with high probability and exhibit robust travel times under variability.

9 Conclusions

This paper introduced a matheuristic for capacitated groundfish survey planning that improves an initial segmented route using repeated, time-limited two-segment MIP refinements. Mandatory port returns and direction-dependent distances make large-scale survey planning computationally difficult. The full capacitated MIP is intractable for operationally sized instances, yet the proposed MH approach reliably improves initial plans across a range of initialization strategies while keeping computation times practical.

The MH systematically refines segment boundaries through short two-segment solves that exploit local structure. Experiments on the 2023 Icelandic groundfish survey show that the method consistently reduces total distance, with the best solution improving the baseline by roughly 270 nm (5.8%), equivalent to about a full day of vessel time. Increasing the time limit beyond 2–3 minutes per two-segment iteration yields no consistent gains and substantially increases runtime, confirming that short, repeated local solves are most effective.

Overall, restricted-MIP local improvement offers a practical, scalable mechanism for constructing high-quality, reproducible survey plans suitable for operational use, strategic planning, and scenario comparison. Future work will focus on expanding the neighborhood structure, extending the method to multi-vessel settings, and incorporating stochastic catch and service-time variability.

Availability. The complete implementation of this study and the 2023 survey data are openly available at <https://github.com/HI-IDN/groundfish-singlevessel-routing>.

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References

1. Boschetti, M.A., Letchford, A.N., Maniezzo, V.: Matheuristics: survey and synthesis. *International Transactions in Operational Research* **30**(6), 2840–2866 (Nov 2023). <https://doi.org/10.1111/itor.13301>
2. Boschetti, M.A., Maniezzo, V.: Matheuristics: using mathematics for heuristic design. *4OR* **20**(2), 173–208 (Jun 2022). <https://doi.org/10.1007/s10288-022-00510-8>
3. David L. Applegate, Robert E. Bixby, Vašek Chvátal, William J. Cook: *The Traveling Salesman Problem: A Computational Study*. Princeton University Press (2011)
4. Dell’Amico, M., Montemanni, R., Novellani, S.: Matheuristic algorithms for the parallel drone scheduling traveling salesman problem. *Annals of Operations Research* **289**(2), 211–226 (Jun 2020). <https://doi.org/10.1007/s10479-020-03562-3>
5. Fan, J., Zhang, C., Yang, F., Shen, W., Gao, L.: A matheuristic with re-lot-sizing strategies for flexible job-shop rescheduling problem with lot-streaming and machine reconfigurations. *European Journal of Operational Research* **319**(3), 747–762 (Dec 2024). <https://doi.org/10.1016/j.ejor.2024.07.030>
6. Gurobi Optimization, LLC: Gurobi Optimization, <https://www.gurobi.com/>
7. Ingibjörg G. Jónsdóttir, Hjalti Karlsson, Höskuldur Björnsson, Jón Sólmundsson, Klara B. Jakobsdóttir, Valur Bogason: A manual for the Icelandic Groundfish Survey in spring 2025. Kver KV 2025-02, Marine and Freshwater Research Institute, Hafnarfjörður, Iceland (Feb 2025), https://www.hafogvatn.is/static/research/files/rallhandbok_2025-enska-1.pdf
8. Jónsdóttir, I.G., Karlsson, H., Pétursson, H., Björnsson, H., Sólmundsson, J., Bogason, V.: Stofnmælingar botnfiska á Íslandsmiðum 2023: Framkvæmd og helstu niðurstöður. Tech. Rep. HV 2023-12, Hafrannsóknastofnun, Hafnarfjörður, Iceland (Mar 2023), https://www.hafogvatn.is/static/research/files/hv_2023_12.pdf
9. Jónsson, M., Rúnarsson, T.P., Steinarsson, B.: Optimizing the Sailing Route for Fixed Groundfish Survey Stations. *International Council for the Exploration of the Sea (ICES)* (1996)
10. Kimura, Daniel K and Somerton, David A: Review of Statistical Aspects of Survey Sampling for Marine Fisheries. *Reviews in Fisheries Science & Aquaculture* **14**(3), 245–283 (2006)
11. Kruskal, J.B.: On the Shortest Spanning Subtree of a Graph and the Traveling Salesman Problem. *Proceedings of the American Mathematical Society* **7**(1), 48–50 (1956). <https://doi.org/10.2307/2033241>
12. Rosenkrantz, D.J., Stearns, R.E., Lewis, II, P.M.: An Analysis of Several Heuristics for the Traveling Salesman Problem. *SIAM Journal on Computing* **6**(3), 563–581 (1977). <https://doi.org/10.1137/0206041>
13. Rúnarsson, T.P.: Interactive Evolutionary Reoptimization for Groundfish Survey Planning. In: Singh, H., Ray, T., Knowles, J., Li, X., Branke, J., Wang, B., Oyama, A. (eds.) *Evolutionary Multi-Criterion Optimization (EMO 2025)*. pp. 275–287. Springer Nature, Singapore (2025). https://doi.org/10.1007/978-981-96-3506-1_19